SmartCampus AI

Comprehensive System Tech Stack, Architecture & Dependency Inventory

Executive Overview: SmartCampus AI is an enterprise-grade autonomous educational intelligence platform. The architecture combines a **Next.js 16 (React 19)** modern web dashboard, a **NestJS 11 (Node.js)** high-throughput API gateway, and an asynchronous **FastAPI / PyTorch** computer vision engine. The platform is powered by state-of-the-art vision models (SCRFD 10G, YOLO26-Face, ArcFace 512D, MediaPipe), an in-memory & persisted **FAISS vector database**, multi-camera spatial trackers, and an automated incremental face enrollment pipeline.

1. Frontend Web & Visualization Tier (apps/web)

Built on modern React 19 and Next.js 16 with a high-performance Glassmorphism design system.

Category / Technology	Version	Architectural Purpose & Implementation
Next.js App Router	v16.2.9	Core frontend framework. Server-Side Rendering (SSR), API rewrites, middleware proxying, and Turbopack bundler.
React & React DOM	v19.2.4	Modern concurrent React engine with Server Components, Actions, and optimized state reconciliation.
TypeScript	v5.x	Strict type system guaranteeing end-to-end type safety across dashboard pages, hooks, and API contracts.
Tailwind CSS	v4.0	Utility-first styling with custom CSS design tokens, responsive breakpoints, and modern dark-mode palettes.
Framer Motion	v12.40.0	Hardware-accelerated fluid micro-animations, modal transitions, and dynamic real-time progress indicators.
Three.js & @react-three/fiber	v0.170 / v9.6	WebGL 3D graphics rendering for the interactive 3D Campus Digital Twin and spatial facility monitoring.
@react-three/drei & postproc	v10.7 / v2.19	Helpers, orbit controls, camera rigging, bloom, and depth-of-field shaders for the campus 3D model.
Lucide React	v1.18.0	Modern clean SVG iconography for biometric dashboards, session states, and navigation modules.
Recharts	v3.8.1	Interactive attendance trends, quality distributions, student risk scoring, and behavior analytics charts.
Zustand & TanStack Virtual	v5.0 / v3.14	Lightweight global state management and high-performance virtualized scrolling for thousands of face crops.
@dnd-kit (Core & Sortable)	v6.3 / v10.0	Drag-and-drop student cluster assignment, manual identity merge, and verification queues.
Socket.io Client	v4.8.3	Real-time dual-way telemetry: live video frames, segment telemetry, and candidate detection alerts.
NextAuth.js & Zod	v4.24 / v4.4	Role-based session management, JWT token handling, and runtime schema validation.

2. Backend API Gateway & Business Logic Tier (apps/api)

Enterprise microservices architecture handling persistence, RBAC security, and device orchestration.

Category / Technology	Version	Architectural Purpose & Implementation
NestJS Core & Common	v11.0.1	Modular Node.js enterprise framework utilizing Dependency Injection, Controllers, Services, and Guards.
Prisma ORM & pg-adapter	v7.8.0	Next-gen type-safe ORM connecting to PostgreSQL 16. Manages migrations, relation mappings, and transactions.
PostgreSQL	v16 Alpine	Primary relational database storing Students, Sessions, Attendance, FaceEmbeddings, Clusters, and Audit Logs.
Passport & JWT (@nestjs/jwt)	v11.0.2	Stateless JSON Web Token authentication with refresh rotation, multi-role RBAC, and secure cookie parsing.
Argon2	v0.44.0	Winner of Password Hashing Competition (PHC); cryptographic hashing for all credentials.
Socket.IO & WebSockets	v4.8.3 / v11.1	Bi-directional real-time communication broadcasting session progress, cluster events, and live alerts.
@nestjs/throttler	v6.5.0	Rate limiting and DDoS mitigation against brute-force attacks on enrollment and attendance endpoints.
Class-Validator & Zod	v0.15 / v4.4	Strict declarative input validation and transformation pipes on all DTO payloads.
RxJS	v7.8.1	Reactive stream composition, event mapping, and asynchronous task management.

3. Computer Vision, Deep Learning & Vector Search Tier (apps/ai-service)

State-of-the-art vision models, biometrics, tracking, and vector indexing running in Python.

Category / Technology	Version	Architectural Purpose & Implementation
FastAPI & Uvicorn	v0.115 / v0.30	High-performance asynchronous Python web server serving REST, live frames, video pipeline, and WebSockets.
YOLO26-Face / Ultralytics	v8.4.116	PyTorch-based ultra-fast face detection model (`yolo26s-face.pt`). Optimized for dense classroom crowds.
SCRFD 10G (InsightFace)	v0.7.3	Sample and Computation Redistribution Face Detector (`det_10g.onnx`). High-recall wide-angle detector.
Combined Dual-Detector	Custom Hybrid	IoU / NMS fusion layer combining SCRFD (high recall) and YOLO26 (dense crowds) into a unified detector.
ArcFace (ResNet-50)	w600k_r50	Additive Angular Margin Loss deep neural network generating 512-dimensional L2-normalized face embeddings.
MediaPipe Landmarker	v0.10.14	468-point 3D facial mesh assessing pitch/yaw/roll head pose, facial symmetry, blur, and lighting quality.
FAISS (Facebook AI Search)	faiss-cpu 1.7	Billion-scale similarity search with IndexFlatIP (inner product / cosine distance). Thread-safe with section isolation.
FaceTracker Engine	Locked Kalman	Multi-object tracker with ByteTrack matching, Kalman filtering, seat anchor stabilization, and feature memory.
HDBSCAN & Agglomerative	v0.8.39 / v1.5	Hierarchical density-based and agglomerative clustering for grouping face tracks into student identities.

Identity Consolidator	Two-Stage Merge	Centroid cosine similarity gating (0.75+), top-k pairwise cross-check, and purity gating preventing false merges.
PyTorch & Torchvision	v2.8.0 / v0.23	Deep learning execution runtime supporting CPU and Apple Silicon Metal acceleration.
ONNX Runtime	v1.19.2	Cross-platform inference accelerator executing SCRFD and ArcFace ONNX models at maximum throughput.
OpenCV & NumPy & SciPy	v4.10 / v1.26	Image transformations, face crop alignment, warp-affine matrix calculation, and Laplacian variance blur scoring.

4. Storage, Infrastructure & Messaging Tier

Distributed containerized services, messaging queues, and object storage backing the platform.

Category / Technology	Version / Port	Architectural Purpose & Implementation
Docker & Docker Compose	v2.x / Multi-Container	Containerized infrastructure with dev/prod profiles managing database, cache, storage, and broker services.
Redis Cache & Pub/Sub	v7.0 Alpine (6379)	In-memory cache for live camera sessions, active track states, rate limiting, and pub/sub message brokering.
MinIO Object Storage	RELEASE (9000/9002)	High-performance S3-compatible cloud object storage for raw classroom footage, face crops, and student avatars.
Eclipse Mosquitto (MQTT)	v2.0 (1883/9001)	Lightweight MQTT publish/subscribe messaging broker for classroom IoT edge camera nodes and status telemetry.
ReportLab PDF Engine	v5.0.0	Enterprise PDF document compilation engine generating programmatic reports, tech stacks, and audit sheets.
GitHub MCP Server	v0.6.2	Model Context Protocol integration for automated version control, issue tracking, and repository monitoring.

5. End-to-End Data Flow & Automated Recognition Pipeline

Stage	Component	Processing Action
1. Video Ingestion	ClassroomPipeline (OpenCV)	Sequential 150s interval segmentation of 1080p footage at dynamic sample FPS (1.5-2.0 FPS).
2. Dual Face Detect	Combined Detector	SCRFD 10G + YOLO26-Face with IoU NMS merging bounding boxes down to ~47 faces/frame.
3. Spatial Tracking	FaceTracker (Kalman)	Continuously associates detections into spatio-temporal tracks with seat-anchor stabilization.
4. Quality & Landmarks	MediaPipe & Laplacian	Filters crops below quality score (<45), estimates pose yaw/pitch/roll, and selects top-quality samples.
5. Embedding Extraction	ArcFace (w600k_r50)	Extracts 512-dimensional L2-normalized facial biometric vectors via ONNX Runtime.
6. Vector Matching	FAISS IndexFlatIP	Cosine search against registered students with strict section whitelist isolation (sim >= 0.75).
7. Profile Adaptation	Continuous Learning	Diversity gating (0.70 <= sim <= 0.95), deduplication, max 12 samples, and PostgreSQL sync.
8. Identity Consolidation	HDBSCAN & Consolidator	Clusters unknown tracks into novel student identities, merges split clusters, and warns on multi-ID.
9. Roll Call & Audit	Prisma & PostgreSQL	Persists attendance records, session cluster status, and audit history; broadcasts live HUD via WS.

Deployment Verification: *All listed dependencies, deep learning checkpoints, database migrations, and container configurations are actively deployed and verified in the SmartCampus AI project environment on macOS / Linux.*